

CDO Risk Report

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Executive Summary

We simulated 1,000 possible five-year outcomes for a \$100 million portfolio of ten corporate bonds. Our results show that equity absorbs most of the cash-flow losses, while the payment priority gives Classes A and B substantial protection. Average collateral receipts are \$115.8324 million against a no-default benchmark of \$130 million. Class A receives less than its scheduled payments in 0.30% of scenarios, compared with 1.10% for Class B.

Our sensitivities show that higher default probability and LGD reduce expected receipts, while higher correlation creates fewer but more severe loss scenarios. At the base settings, collateral VaR95 is \$41.2800 million, and ES95 is \$53.3430 million. These results depend on one-time recovery payments and tranche principal being due only at maturity. We analyze cash-flow risk here; valuation comes later.

Key Model Assumptions

Key factor	Summary
Portfolio and simulation	1,000 scenarios for ten bonds with \$10 million face value each. Bonds mature in five years and pay 6% annual coupons quarterly.
Credit assumptions	4% annual default probability, 60% loss given default, and 20% target population correlation between default times.
No-default benchmark	\$130.0000 million in total collateral receipts: \$100 million principal plus \$30 million coupons.
Expected collateral loss	\$14.1677 million relative to the no-default benchmark, implying average receipts of \$115.8324 million.
Payment priority and protection	Cash is allocated to A, then B, then equity. A experiences shortfalls in 3 scenarios (0.30%); B in 11 scenarios (1.10%).
Recovery treatment	Each defaulted bond pays \$4 million once, at the end of its default quarter, with no coupon that quarter and no subsequent payments.
Tranche principal and cash allocation	A and B repay principal only at maturity. Before maturity, recovery cash remaining after scheduled tranche coupons goes to equity. No cash or unpaid shortfalls carry forward.

Methodology

First, we define the portfolio and the tranches: ten bonds with a \$10 million face value each, and Classes A and B with respective notional amounts of \$20 million and \$10 million. Each bond pays a 6% annual coupon, or \$150,000 quarterly, and has a 4% annual default probability. LGD is 60%, implying a \$4 million recovery per defaulted bond. Classes A and B have annual coupons of 2% and 4%, respectively,

so each receives a scheduled quarterly coupon of \$100,000. The waterfall pays A first, then B, and finally equity. We assume that A and B repay their principal entirely in quarter 20.

We generate 1,000 scenarios using a fixed random seed and standardize each normal-draw column to sample mean zero and sample standard deviation one. To target a population default-time correlation of 0.20, we use Gauss–Hermite quadrature to evaluate the correlation after transformation and Brent’s method to calibrate the normal correlation to approximately 0.2332. We then use Cholesky decomposition to correlate all ten normal draws in each scenario, transform them into uniform percentiles, and apply the exponential inverse CDF. The resulting default times have approximately exponential marginals after sample moment matching; sample correlations can differ from the population target.

A defaulted bond pays 40% of face value once, at the end of its default quarter, with no coupon that quarter and no later payments. Surviving bonds continue to pay coupons and repay principal at maturity. Each quarter, we pool coupons, recoveries, and principal and apply the waterfall. Recovery cash remaining after scheduled tranche payments goes to equity; there are no reserves, early tranche principal repayments, or carry-forwards.

Statistical Analysis

For each scenario, we sum the 20 quarterly payments for collateral, A, B, and equity. We then calculate statistics across the 1,000 scenarios.

Table 1. Five-Year Cash Receipts: Portfolio and Tranche Summary
USD millions | 1,000 simulated scenarios | Undiscounted

Statistic	Collateral portfolio	Class A	Class B	Residual equity
No-default benchmark	130.0000	22.0000	12.0000	96.0000
Mean receipts	115.8324	21.9702	11.9022	81.9601
Standard deviation	14.3977	0.5444	0.9482	13.9721
Minimum observed	54.1000	12.0500	1.6000	34.0500
5th percentile	88.7200	22.0000	12.0000	54.7200
Median	121.4500	22.0000	12.0000	87.4500
95th percentile	130.0000	22.0000	12.0000	96.0000
Maximum observed	130.0000	22.0000	12.0000	96.0000

Notes: Results assume a 4% annual default probability, 60% LGD, and a 0.20 target default time correlation.

Across 1,000 simulated scenarios, the collateral portfolio generates average five-year cash receipts of \$115.83 million, compared with the \$130 million no-default benchmark. Equity absorbs most of the reduction, receiving an average of \$81.96 million against its \$96 million benchmark. Classes A and B

receive close to their full scheduled amounts on average because of their payment priority. Their shortfalls occur in fewer than 5% of scenarios, so their 5th-percentile receipts still equal their scheduled totals. The minimum values describe the worst observed cases.

Table 2. Five-Year Expected Loss and Probability of Loss
USD millions | 1,000 simulated scenarios | Undiscounted

Metric	Collateral portfolio	Class A	Class B	Residual equity
Expected loss (USD millions)	14.1677	0.0299	0.0979	14.0400
Probability of loss	72.20%	0.30%	1.10%	72.20%

Note: Loss equals no-default (benchmark) minus actual receipts. Expected loss averages all scenarios, including those with no loss. Probability of loss is the percentage of scenarios with a loss greater than one cent.

Expected collateral loss is \$14.1677 million. Expected losses for A, B, and equity are \$29,850, \$97,850, and \$14.0400 million, respectively. These results show that equity absorbs most of the cash-flow reduction. Loss frequencies are 72.20% for collateral, 0.30% for A, 1.10% for B, and 72.20% for equity. Class A's payment priority makes a shortfall less frequent than for Class B. In this simulation, every scenario with a collateral loss also reduces equity receipts.

Table 3. Five-Year Value at Risk and Expected Shortfall
USD millions | 95% confidence level | 1,000 simulated scenarios | Undiscounted

Risk measure	Collateral portfolio	Class A	Class B	Residual equity
Value at Risk (VaR95)	41.2800	0.0000	0.0000	41.2800
Expected Shortfall (ES95)	53.3430	0.5970	1.9570	50.7890

Note: Loss is measured relative to no-default cash receipts.

VaR95 is the interpolated 95th percentile of five-year cash-flow loss. It is \$41.2800 million for both collateral and equity, and zero for A and B. ES95 averages the 50 largest losses for each class: \$53.3430 million for collateral, \$0.5970 million for A, \$1.9570 million for B, and \$50.7890 million for equity. Zero VaR95 does not mean that A and B cannot lose cash; their positive ES95 captures losses in the small number of shortfall scenarios. Tail measures are calculated separately for each class.

Outcome Distributions

Table 4. Probability Mass Function of Five-Year Bond Defaults
10-bond portfolio | 1,000 simulated scenarios

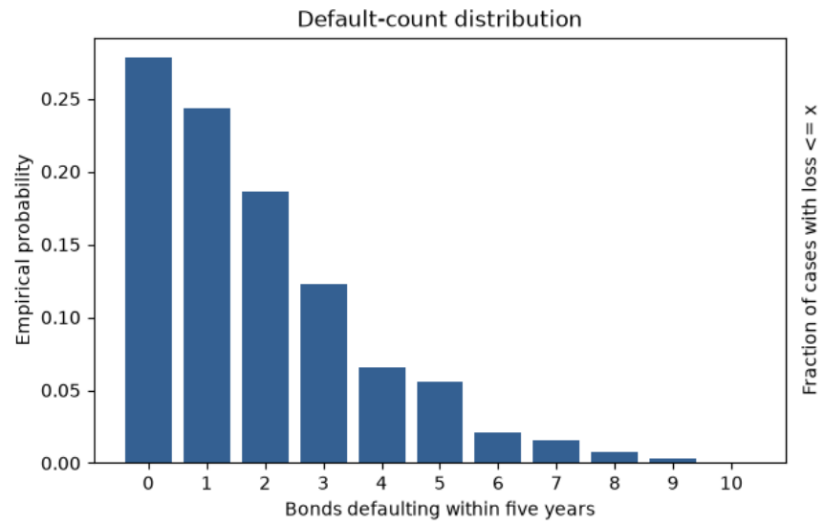
Number of defaulted bonds	Number of scenarios	Empirical probability
0	278	27.80%
1	243	24.30%
2	186	18.60%
3	123	12.30%
4	66	6.60%
5	56	5.60%
6	21	2.10%
7	16	1.60%
8	8	0.80%
9	3	0.30%
10	0	0.00%
Total	1,000	100.00%

Note: Each probability equals the number of scenarios with exactly the indicated default count divided by the total number of cases (1,000).

Scenarios with fewer defaults are more common. No defaults occur in 27.80% of cases, while at least one default occurs in 72.20%. The frequency generally declines as the default count rises. We observe no case with all ten bonds defaulting, but this does not mean that such an outcome is impossible.

Graph 1. Default Count Distribution

10-bond portfolio | 1,000 simulated scenarios

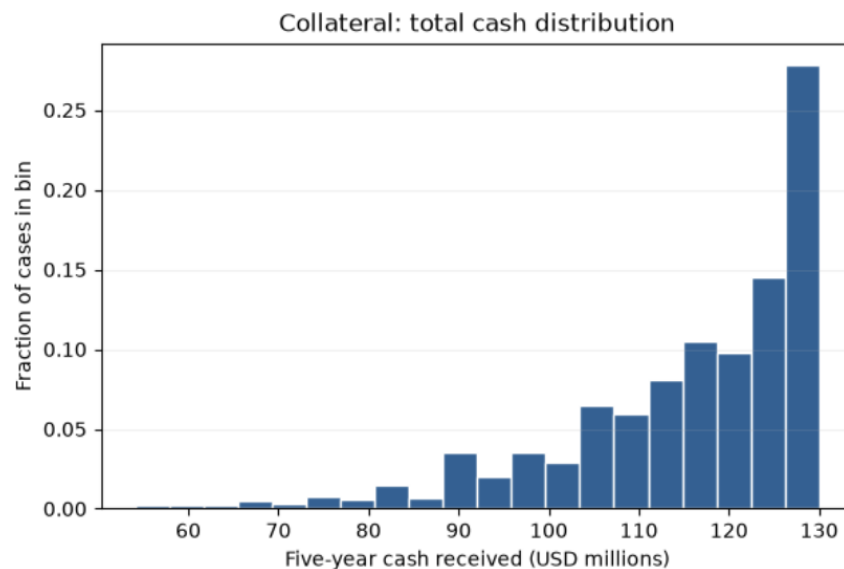


Note: Each bar shows the simulated probability of exactly the indicated number of bonds defaulting within five years.

The graph illustrates the PMF in Table 4. This default-count distribution is estimated from correlated default times. The underlying model uses exponential default-time marginals; the number of defaults has a discrete distribution over zero to ten and does not follow an exponential distribution.

Graph 2. Total Cash Distribution

10-bond portfolio | 1,000 simulated scenarios



Note: The histogram shows collateral cash receipts over five years. Each bar represents the proportion of scenarios within a cash-receipt range.

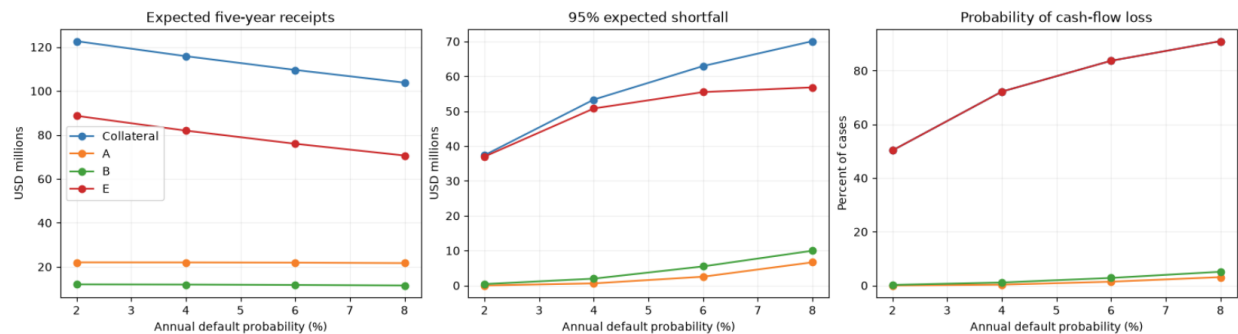
Cash receipts are concentrated near the no-default benchmark. The distribution is left-skewed, with a probability mass at the maximum receipt of \$130 million and a lower tail reflecting principal losses and missed coupons.

For the graphs of the collateral and equity loss CDF, please see the appendix.

Sensitivity Analysis

Annual Default Probability Sensitivity

Graph 3. Default Probability Sensitivity
10-bond portfolio | 1,000 simulated scenarios

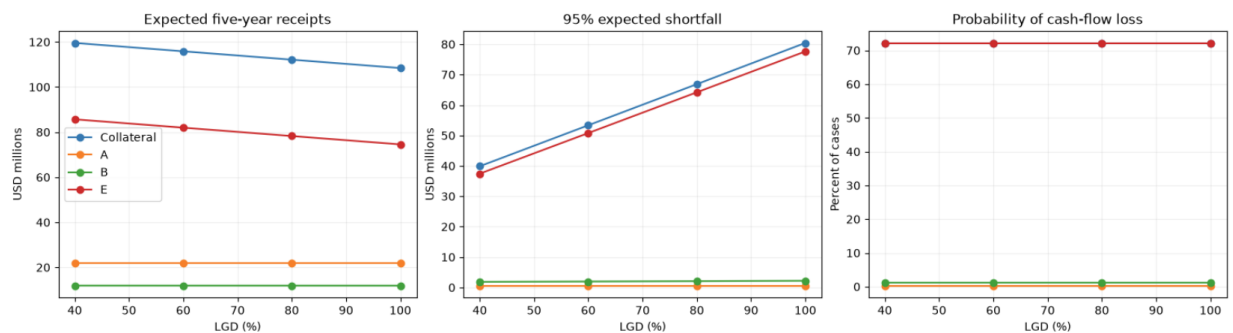


Note: Annual default probability varies from 2% to 8%, holding LGD at 60% and target default-time correlation at 0.20, using the same underlying random draws.

Higher annual default probability reduces expected collateral and equity receipts and increases losses in the worst 5% of scenarios. A and B are less sensitive in average-dollar terms, but they are not unaffected: their shortfall frequencies and tail losses also increase. The waterfall gives them payment priority, while equity bears most of the reduction in receipts.

LGD Sensitivity

Graph 4. LGD Sensitivity
10-bond portfolio | 1,000 simulated scenarios

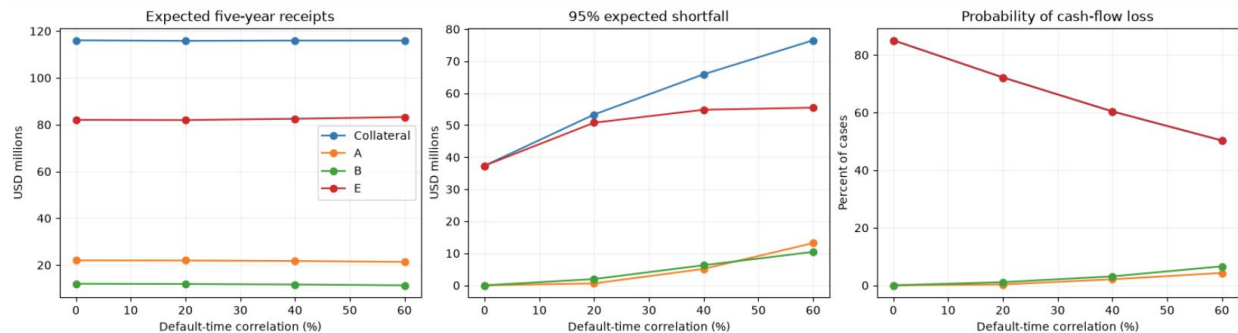


Note: LGD varies from 40% to 100%, holding annual default probability at 4% and target default-time correlation at 0.20, using the same underlying random draws.

Higher LGD reduces recovery payments, lowering expected collateral and equity receipts and increasing ES95. Their loss frequencies remain unchanged because the same default times are used in these runs. A and B show comparatively limited sensitivity over the tested range, but their payment priority does not guarantee full payment in every scenario.

Default Time Correlation Sensitivity

Graph 5. Default-Time Sensitivity
10-bond portfolio | 1,000 simulated scenarios



Note: Target default-time correlation varies from 0% to 60%, holding annual default probability at 4% and LGD at 60%, using the same underlying random draws.

Higher correlation concentrates defaults into fewer, more severe scenarios. Collateral and equity losses therefore occur less frequently, while their average losses in the worst 5% of scenarios increase. A and B experience more frequent shortfalls. Expected collateral receipts remain approximately unchanged because individual default probabilities are fixed. A lower frequency of equity loss should therefore not be interpreted as lower tail risk.

Conclusion

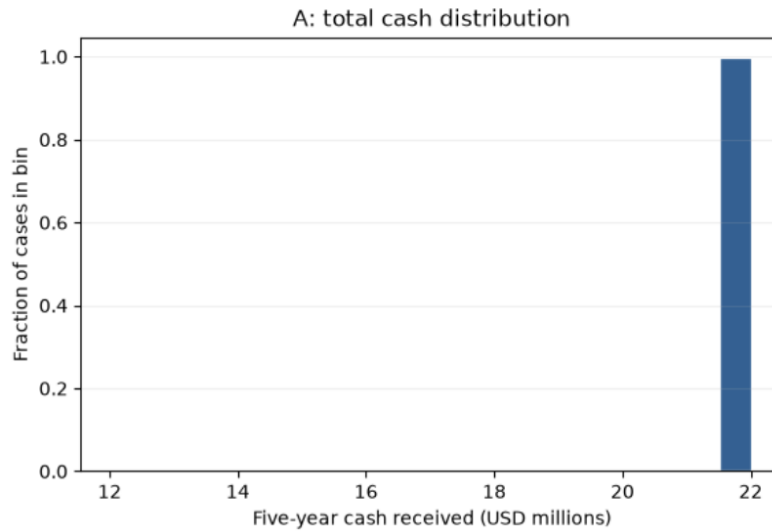
Our results show how the waterfall allocates the same portfolio risk differently across investors. A receives the greatest protection, followed by B, while equity absorbs most of the cash-flow reduction. However, the rare shortfalls in A and B and their positive expected shortfall show that payment priority does not eliminate risk.

The sensitivity analysis shows that higher default probability and LGD reduce expected collateral receipts, while higher correlation mainly changes the concentration and severity of losses. Recovery timing also matters: cash distributed to equity before maturity cannot cover later tranche principal shortfalls.

Appendix

Graph A.1. A Total Cash Distribution

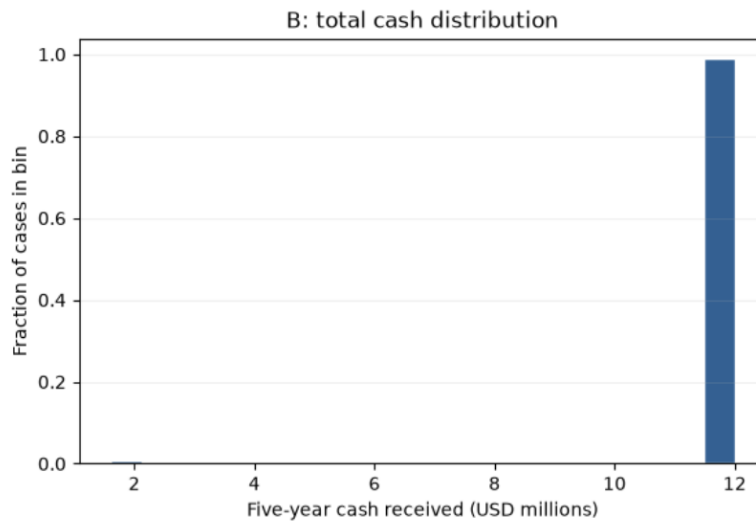
10-bond portfolio | 1,000 simulated scenarios



Note: Class A receives its full scheduled \$22 million in 99.7% of scenarios. Three scenarios produce payment shortfalls, forming a small lower tail that is difficult to see at this scale. Its senior payment priority provides substantial protection under the modeled assumptions.

Graph A.2. B Total Cash Distribution

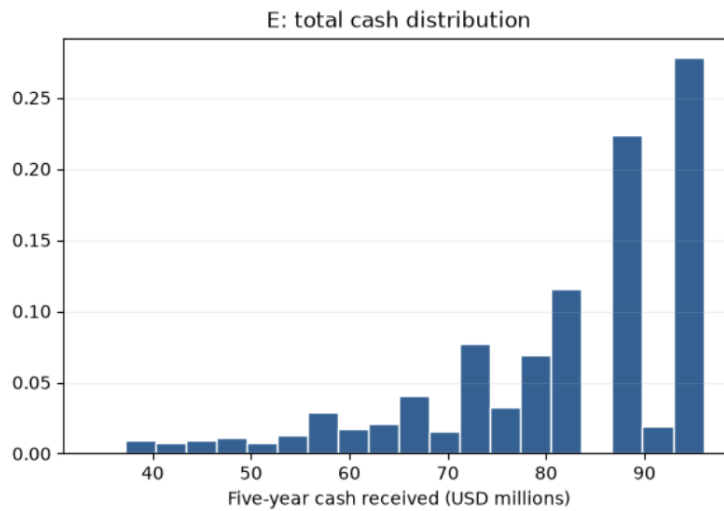
10-bond portfolio | 1,000 simulated scenarios



Note: Class B receives its full scheduled \$12 million in 98.9% of scenarios. Eleven scenarios produce payment shortfalls. Because Class B is paid after Class A, it experiences shortfalls more frequently.

Graph A.3. E Total Cash Distribution

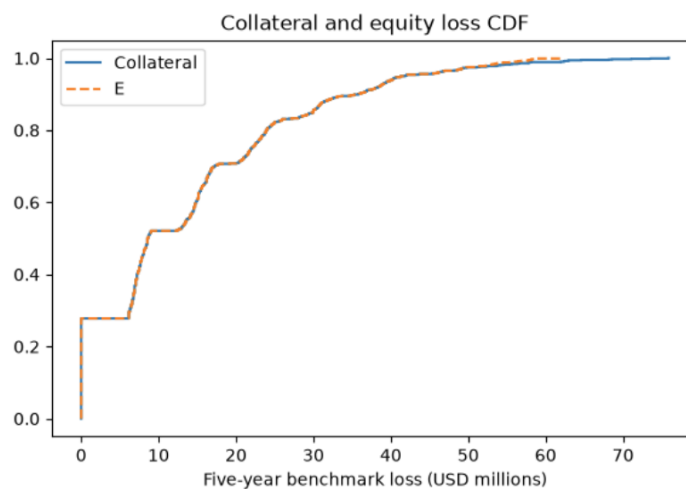
10-bond portfolio | 1,000 simulated scenarios



Note: Equity receives the residual cash after scheduled payments to Classes A and B. Its five-year receipts average \$81.96 million, compared with \$96 million without defaults. The broad distribution and lower tail reflect equity's exposure to collateral losses. These amounts are cash receipts, not investment profits or returns.

Graph A.4. Collateral and Equity Loss CDF

10-bond portfolio | 1,000 simulated scenarios



Note: Each curve shows the fraction of scenarios with five-year cash-flow losses at or below the amount on the horizontal axis. Losses are measured against each class's no-default benchmark. The curves largely overlap because equity absorbs most collateral losses; they diverge in severe scenarios when Classes A and B also experience shortfalls.